For each class, briefly describe the architecture, hyperparameters, and optimization strategy used. Report the validation performance of all models on identical train/validation splits.

2 Data-Driven Discovery of a Discrete-Time Recurrence [20 Marks]

2.1 Dataset and Problem Setup [3 Marks]

You are provided with multiple univariate sequences $\{x_k\}_{k=0}^{T}$ generated by an **unknown, time-invariant, autonomous, discrete-time mechanism**. The data are noisily observed but follow a fixed rule within each sequence. Your goals are to (i) build predictors of x_k from its history, (ii) *identify* a compact analytical recurrence consistent with the data (including its effective order), and (iii) analyze how well it fits to the data. You are suggested to use an RNN for this task. Dataset is at Dataset link.

Tasks

1. **Data splits:** Create chronological train/validation/test splits. Justify lengths.
2. **Supervised pairs:** Form $(\mathbf{h}_k \mapsto x_k)$ with history vectors of length p and explain how p is selected.
3. **Normalization:** Apply normalization based on the training set only; describe inverse transform.

Suggested Implementation Skeleton

- `data_prep.py`: load data, create time splits, and normalize.
- `model.py`: implement baseline predictors (e.g., Linear AR, MLP, or small RNN).
- `train.py`: train chosen models, tune p, and log results.
- `identify.py`: derive analytical recurrence F_θ consistent with learned mapping.
- `analyze.py`: plot residuals, and report test performance.

2.2 Sequence Prediction [6 Marks]

Train one or more predictors mapping a chosen history to the next value:

$$\hat{x}_k = g_\phi\big(x_{k-1}, x_{k-2}, \ldots, x_{k-p}\big),$$

and later use this to recover a *recurrent* analytical expression. (Hint : *recurrent*)

Suggested Deliverables

- Trained models and hyperparameter table.
- Prediction plots and validation curves.
- Short discussion on history length and model complexity.

2.3 Analytical Recurrence Identification [6 Marks]

Identify a closed-form recurrence

$$\hat{x}_k = F_\theta\big(x_{k-1}, x_{k-2}, \ldots, x_{k-\hat{p}}\big),$$

and compare residuals with black-box predictors.

2.4 Evaluation Criterion

- With MAE, MSE as your metrics, evaluate on single-step prediction with your DL model.
- Evaluate autoregressive generation, both with your DL model, and the analytical expression that you've fit to your data.
- Show variation in error with these models, with increasing forecasting length.

2.5 Parsimony and Stability [3 Marks]

Explore parameter count vs. performance. Select the simplest accurate model and present as to how well it fits to the data in comparison to other sweeps. What does this tell you about the dataset ? Is there any temporal relation you can extract from the dataset.

Report Checklist

- Model specifications, identified F_θ, parameter estimates.
- Complexity–accuracy trade-off figure.
- Stability plots and conclusions.
